

All JavaScript Array Functions

```
let arr = [1, 2, 3, 4, 5];  
let arr2 = ["a", "b", "c"];
```

Add / Remove

Push: Adds element(s) at the end, returns new length.

Pop: Removes the last element, returns it.

Shift: Removes the first element, returns it.

Unshift: Adds element(s) at the start, returns new length.

Splice: Adds, removes, or replaces elements at a position.

Slice: Copies part of the array into a new one.

Concat: Merges arrays into a new one.

```
arr.push(6);           // [1,2,3,4,5,6] -> returns 6 (new length)  
arr.pop();             // [1,2,3,4,5] -> returns 6  
arr.shift();           // [2,3,4,5] -> returns 1  
arr.unshift(0);        // [0,2,3,4,5] -> returns 5 (new length)  
arr.splice(2, 1, 99);   // [0,2,99,4,5] -> returns [3]  
arr.slice(1, 3);        // [2,99]  
arr.concat(arr2);       // [0,2,99,4,5,"a","b","c"]
```

Search

indexOf: Finds index of first occurrence.

lastIndexOf: Finds index of last occurrence.

Includes: Checks if value exists in the array.

Find: Returns first element matching a condition.

findIndex: Returns index of first match.

Filter: Returns all elements that match a condition.

```
arr.indexOf(99);        // 2  
arr.lastIndexOf(5);     // 4  
arr.includes(4);        // true  
arr.find(x => x > 3);    // 99  
arr.findIndex(x => x > 3); // 2  
arr.filter(x => x > 2);  // [99,4,5]
```

Conversion / Info

toString: Converts array to a comma-separated string.

join: Joins array elements into a string with a custom separator.

```
arr.toString();           // "1,2,3,4,5"  
arr.join("-");           // "1-2-3-4-5"
```

Iteration / Mapping

forEach: Runs a function for each element (no return).

Map: Creates a new array by transforming each element.

flatMap: Map + flatten one level in one step.

```
arr.forEach(x => x * 2);   // runs fn, no return (undefined)  
arr.map(x => x * 2);       // [0,4,198,8,10]  
arr.flatMap(x => [x, x*2]); // [0,0,2,4,99,198,4,8,5,10]
```

Reduce / Accumulate

Reduce: Reduces array to a single value (left → right).

reduceRight: Reduces array to a single value (right → left).

```
arr.reduce((a,b)=>a+b,0); // 110  
arr.reduceRight((a,b)=>a+b,0); // 110
```

Sorting / Reversing

Sort: Sorts the array in place.

Reverse: Reverses array order in place.

```
arr.sort();               // [0,2,4,5,99]  
arr.reverse();            // [99,5,4,2,0]
```

Copy / Fill

copyWithin: Copies values inside array to another index.

Fill: Fills array with a static value.

```
arr.copyWithin(0, 2, 4);    // [4,2,4,2,0]
arr.fill(1, 1, 3);         // [4,1,1,2,0]
```

Flattening

Flat: Flattens nested arrays one level deep.

```
[[1],[2,3]].flat();        // [1,2,3]
```

String conversion

toLocaleString: Localized string representation of array.

```
arr.toLocaleString();      // "4,1,1,2,0"
```

Iterators

Keys: Iterator of indexes.

Values: Iterator of values.

Entries: Iterator of [index, value] pairs.

```
arr.keys().next().value;    // 0 (first key)
arr.values().next().value;  // 4 (first value)
arr.entries().next().value; // [0,4]
```

Utility

Array.isArray: Checks if a value is an array.

```
Array.isArray(arr);        // true
```

Static creation helpers

Array.from: Creates array from iterable/array-like.

Array.of: Creates array from given arguments.

```
Array.from("123");           // ["1","2","3"]
Array.of(7,8,9);             // [7,8,9]
```

Newer goodies

at: Gets element by index (supports negatives).

findLast: Returns last element matching a condition.

findLastIndex: Returns index of last match.

```
arr.at(-1);                   // 0 (last element)
arr.findLast(x => x < 5);     // 2
arr.findLastIndex(x => x < 5); // 3
```